Coalitional monotonicity and the core: minimal impossibility, an exact-games escape, and the boundary of surplus-distributor solutions

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Abstract

We study coalitional monotonicity of single-valued core selections in transferable-utility (TU) games. First, we pin down the minimal size of an impossibility certificate for the classical incompatibility of core selection and coalitional monotonicity: three-game certificates (a base game and two single-coalition increases) appear already in the textbook proof of Peleg and Sudhölter, and we record the complementary elementary fact that two-game families are always feasible—so three is exactly minimal—together with a self-contained three-game certificate admitting a six-line proof. Second, we map the monotonicity boundary of the Arin–Katsev surplus-distributor (SD) prenucleolus: coalitional monotonicity fails already on *exact* games ($n = 4$, a minimized witness), the SD-prekernel is multi-valued on exact games, population monotonicity fails even on convex games ($n = 3$, minimal), and on permutation games the SD solution amplifies the nucleolus's failure of pairwise monotonicity, whereas on assignment games extensive searches support monotonicity of both solutions. Third, we study whether the impossibility descends to exact games. For $n \leq 4$ it does not: by known results on large cores and egalitarian solutions, a coalitionally monotonic core selection exists on four-player exact games. We then prove by an exact positive-circuit classification that every three-game star of exact games is feasible through five players: the two genuinely open five-player arm geometries comprise 609 symmetry orbits (5,670 labeled circuit supports), all neutralized by balancedness, bump nonnegativity, or endpoint exactness. The Housman–Clark mechanism is provably unavailable on exact games of any size, common-core intersection of bumped exact games can nevertheless fail (a four-arm example whose family remains feasible), and by a compactness argument existence for each n is *equivalent* to feasibility of all finite bump families, which we conjecture for $n \geq 5$.

Keywords: cooperative games; core; coalitional monotonicity; nucleolus; SD-prenucleolus; exact games.

1 Introduction

A transferable-utility (TU) cooperative game assigns to each coalition the worth it can guarantee on its own; a single-valued solution recommends one division of $v(N)$. Two desiderata have

*A machine-readable certificate supplement with a standalone one-command verifier accompanies this paper as Online Resource 1; see Section 8 for the verification methodology and the disclosure of AI assistance.

organized much of the literature on such solutions. *Core selection*: the recommendation should be stable against coalitional deviation whenever stability is possible. *Coalitional monotonicity* (Young 1985): if exactly one coalition's worth rises, no member of that coalition should receive less.

The two desiderata are incompatible. Young (1985) proved that for $|N| \geq 5$ no core selection is coalitionally monotonic; Housman and Clark (1998) extended the impossibility to $|N| \geq 4$ and showed it is tight ($n = 3$ admits an infinite family of monotonic core selections). The nucleolus fails coalitional monotonicity even on four-player convex games (Hokari 2000), as do the per-capita nucleolus (Arin 2013) and the core-center (Mirás Calvo et al. 2021). Against this background, Arin and Katsev (2011, 2014, 2016a, 2018) introduced the *surplus-distributor prenucleolus* (SD-prenucleolus),[1] a lexicographic core selection that *is* coalitionally monotonic on convex games and on monotonic games with veto players. On the domain of convex games alone, egalitarian solutions in the tradition of Dutta and Ray (1989) are also coalitionally monotonic core selections (Dietzenbacher 2021); what distinguishes the SD-prenucleolus is that it selects from the core on *all* balanced games while retaining monotonicity on both the convex and the monotonic-veto domains. How far that good behavior extends beyond these two classes was, to our knowledge, open.

This paper answers that question and, in doing so, isolates what we believe is the right open problem in this area. Our contributions:

1. **Minimal impossibility (Section 3).** Three-game impossibility families (a base and two bumps) are already present in the literature: the textbook proof of Peleg and Sudhölter (2003, Theorem 8.2.3) has exactly this shape, and the contradiction in the construction of Arin (2013) likewise uses only a base and two of its bumps. What appears not to have been recorded is optimality: we prove that two-game families are always feasible, so three is *exactly* the minimal size of an impossibility certificate, and we give a self-contained three-game certificate whose infeasibility admits a six-line proof.

2. **The SD boundary (Sections 4–5).** The coalitional monotonicity of the SD-prenucleolus does not extend one step beyond convexity: it fails on *exact* games at $n = 4$ (convex $\subsetneq$ exact $\subsetneq$ totally balanced $\subsetneq$ balanced), with a minimized integer witness. Three companion results complete the map: the SD-prekernel is multi-valued on exact games (containing a positive-dimensional core segment, so the convex-domain uniqueness theorem of Arin and Katsev (2018) is tight); the SD-prenucleolus is *not* population monotonic even on convex games (witness at the minimal size $n = 3$, with a short hand proof); and on permutation games the SD-prenucleolus fails pairwise monotonicity on the same witness as the classical nucleolus, with a larger violation. Certificates for the computational theorems appear in Appendix A.

3. **The exact-games question (Section 7).** The Young/Housman–Clark impossibility does not descend to exact games with at most four players: combining the large-core result of Biswas et al. (1999) with Dietzenbacher's monotonic egalitarian core selection on PES-stable games (Dietzenbacher 2021) yields a coalitionally monotonic core selection on $\mathcal{E}_4$ (Proposition 7.29). For general n the question is open, and we show the known impossibility mechanisms cannot decide it: the singleton-core pinning mechanism cannot exist

[1] The literature uses both "SD-nucleolus" (e.g. the survey of Iñarra et al. 2020) and "SD-prenucleolus" (the titles of Arin and Katsev 2011, 2014) for the Arin–Katsev construction; we follow the latter. See Section 4.

on exact games (Lemma 7.1); face bounds of groups are tight and invariant along monotonicity chains of proper bumps (Theorem 7.2); overlapping two-arm families always admit a common core point (Lemma 7.3); yet common-core intersection can fail at $m = n = 4$ arms (Proposition 7.26) while the associated family remains feasible—so family feasibility, not core intersection, is the right invariant. Exact bump edges contain whole exact one-parameter arcs (Lemma 2.3), and every minimal infeasible family must be a connected acyclic labeled graph with a sparse Helly–Farkas witness (Proposition 7.28). A compactness argument (Proposition 7.27) shows that for each n, existence of a coalitionally monotonic core selection on exact games is *equivalent* to the feasibility of all finite bump families, which we conjecture for $n \geq 5$ (Conjecture 7.30). The procedural egalitarian route cannot establish this conjecture on all exact games: Mauri Masdeu (2017, Example 5) already gives an exact five-player game that is not PES-stable. At the minimal open size, however, an exhaustive exact classification proves that every base-plus-two-bump family is feasible (Proposition 7.20).

4. **Market games (Section 6).** The nucleolus is *not* pairwise monotonic on permutation games: raising a single entry of the underlying value matrix can strictly lower the payoff of an agent named by that entry (smallest witness found: $n = 4$, integer matrix of total mass 20). In contrast, on assignment games—the two-sided special case—we found no violation for the nucleolus or the SD-prenucleolus in an extensive float-screened search that includes one fully exact component, an exhaustive certified sweep of a 2×2 grid. For the nucleolus this independently supports a theorem announced in the 2008 OTKA project report of Solymosi, Forgó and Pintér (Solymosi et al. 2008), whose proof we have not been able to access; for the SD-prenucleolus the statement appears new and we pose it as a conjecture. The permutation witness shows any proof must use two-sidedness itself.

All positive computational claims in this paper (witnesses, solution values, class memberships) are certified in exact rational arithmetic; lexicographic solutions are certified by Kohlberg-type balancedness criteria and, for headline witnesses, re-derived by an algorithmically independent exact solver. Negative search outcomes are float-screened, as detailed in Section 8; Appendix A lists human-checkable certificates, reproducible with the supplement's standalone verifier.

2 Preliminaries

A *TU game* is a pair (N, v) with $N = \{1, \dots, n\}$ and $v : 2^N \to \mathbb{R}$, $v(\emptyset) = 0$. For $x \in \mathbb{R}^N$ and $S \subseteq N$ write $x(S) = \sum_{i \in S} x_i$. The *core* is $\mathrm{Core}(v) = \{x : \ x(N) = v(N), \ x(S) \geq v(S) \ \forall S\}$; v is *balanced* if $\mathrm{Core}(v) \neq \emptyset$, *totally balanced* if every subgame is balanced, *exact* (Schmeidler 1972) if for every S there is $x \in \mathrm{Core}(v)$ with $x(S) = v(S)$, and *convex* if $v(S \cup T) + v(S \cap T) \geq v(S) + v(T)$ for all S, T. It is standard that convex $\Rightarrow$ exact $\Rightarrow$ totally balanced $\Rightarrow$ balanced, all inclusions strict for $n \geq 4$ (respectively $n \geq 3$ for the last).

A collection $\mathcal{B}$ of nonempty proper coalitions is *balanced* if there are strictly positive weights $(\lambda_S)_{S \in \mathcal{B}}$ with $\sum_{S \in \mathcal{B}} \lambda_S \mathbf{1}_S = \mathbf{1}_N$. A *core selection* on a class $\mathcal{C}$ of balanced games is a map θ with $\theta(v) \in \mathrm{Core}(v)$ for all $v \in \mathcal{C}$.

Definition 2.1 (Coalitional monotonicity (Young 1985; Housman and Clark 1998))**.** *A single-valued solution θ is* coalitionally monotonic *on $\mathcal{C}$ if for all $u, v \in \mathcal{C}$ with $u(T) \leq v(T)$ for some coalition T and $u(S) = v(S)$ for all $S \neq T$, we have $\theta_i(u) \leq \theta_i(v)$ for all $i \in T$.*

Aggregate monotonicity is the special case $T = N$. *Population monotonicity* on a class closed under subgames requires $\theta_i(v|_S) \le \theta_i(v|_T)$ for all $i \in S \subseteq T$; it suffices to check $|T \setminus S| = 1$.

Throughout, a *bump* of v at (S, d), $d > 0$, with $\emptyset \neq S \subseteq N$, is the game $v + d \cdot e_S$ agreeing with v everywhere except $(v + d \cdot e_S)(S) = v(S) + d$; the bump is *proper* if $S \neq N$. A *(bump) family* is a finite set of games together with bump relations between them; a family is *feasible* if one can choose a core element for each game satisfying all the monotonicity inequalities of Definition 2.1 along the bump relations. The existence of a monotonic core selection on a class immediately implies every family inside the class is feasible; Proposition 7.27 states the converse.

We will use repeatedly the following elementary consequence of exactness.

Lemma 2.2. *Let v be exact and $T \subseteq N$ nonempty. Then $\max_{x \in \mathrm{Core}(v)} x(T) = v(N) - v(N \setminus T)$. Moreover, for any partition $\{T_1, \dots, T_m\}$ of N with $m \ge 2$, $\sum_k \big(v(N) - v(N \setminus T_k)\big) \ge v(N)$; for $m = 1$ the same inequality holds trivially with equality.*

Proof. For $x \in \mathrm{Core}(v)$, $x(T) = v(N) - x(N \setminus T)$, so maximizing $x(T)$ is minimizing $x(N \setminus T)$; by exactness the minimum equals $v(N \setminus T)$. For the second claim with $m \ge 2$, the collection $\{N \setminus T_k\}_k$ is balanced with weights $1/(m-1)$, so balancedness of v gives $\sum_k v(N \setminus T_k) \le (m-1)\, v(N)$, which rearranges to the claim. □

Lemma 2.3 (exact bump arcs)**.** *Let v be exact and let S be nonempty. If $S \subsetneq N$ and $v' = v + d e_S$ is exact for some $d > 0$, then $v + t e_S$ is exact for every $0 \le t \le d$. If $S = N$, then $v + t e_N$ is exact for every $t \ge 0$.*

Proof. First let $S \subsetneq N$ and write $C_t = \mathrm{Core}(v) \cap \{x : x(S) \ge v(S) + t\}$. This is the core of $v + t e_S$ and contains $\mathrm{Core}(v')$. Exactness of v supplies a point of its core with $x(S) = v(S)$, while any point of $\mathrm{Core}(v')$ has $x(S) \ge v(S) + d$. Convexity therefore gives a point of C_t tight at S. For $T \neq S$, the function

$$f_T(t) = \min_{x \in C_t} x(T)$$

is nondecreasing, is at least $v(T)$, and equals $v(T)$ at both endpoints by exactness of v and v'. Hence it is identically $v(T)$.

Now let $S = N$. For each proper T, choose $x^T \in \mathrm{Core}(v)$ with $x^T(T) = v(T)$ and a player $j \notin T$. Then $x^T + t e_j$ belongs to the core of $v + t e_N$ and remains tight at T. Efficiency handles N itself. □

Thus every bump edge inside the exact class represents a whole exact one-parameter arc, not merely a relation between two isolated games. This permits local or parametric selection arguments to be integrated along the same edges that define the finite-family feasibility problem.

3 Minimal impossibility on balanced games

Housman and Clark (1998) prove that no core selection on four-player balanced games is coalitionally monotonic, using a family of five games (a base and four bumps). Certificates with three games are in fact already present in the literature: the textbook proof of Peleg and Sudhölter (2003, Theorem 8.2.3) uses precisely a four-player base game with two single-coalition increases, and the contradiction in the construction of Arin (2013) also needs only a base and two of its bumps. The observation that appears to be new here is the matching lower bound: three games cannot be improved upon.

Lemma 3.1. *Every two-game family $\{v,\, v + d \cdot e_S\}$ of balanced games is feasible.*

Proof. If $S \neq N$, then $\mathrm{Core}(v + d \cdot e_S) \subseteq \mathrm{Core}(v)$, since the bump only tightens the constraint at S; pick any $x \in \mathrm{Core}(v + d \cdot e_S)$ and assign it to both games. If $S = N$, pick $x \in \mathrm{Core}(v)$ and assign $x + \frac{d}{n}\mathbf{1}$ to the bumped game: it is efficient for it, satisfies every core constraint (each $x(T)$ only increases), and dominates x coordinatewise. In both cases all monotonicity inequalities hold. □

Theorem 3.2. *Let $N = \{1,2,3,4\}$ and define the balanced game v by*

$$v(\{1,3\}) = v(\{2,3\}) = v(\{1,4\}) = v(\{2,4\}) = v(\{3,4\}) = 3,$$

$$v(\{1,2,3\}) = v(\{1,2,4\}) = v(\{1,3,4\}) = 1, \quad v(\{2,3,4\}) = 5, \quad v(N) = 6,$$

all other coalitions 0. *Let $v_A = v + 3 \cdot e_{\{1,2,3\}}$ and $v_B = v + 1 \cdot e_{\{2,3,4\}}$; all three games are balanced, as witnessed by $(1,1,2,2) \in \mathrm{Core}(v) \cap \mathrm{Core}(v_A)$ and $(0,0,3,3) \in \mathrm{Core}(v_B)$. Then the family $\{v, v_A, v_B\}$ is infeasible: no choice of core elements $x \in \mathrm{Core}(v)$, $y \in \mathrm{Core}(v_A)$, $z \in \mathrm{Core}(v_B)$ satisfies $x_i \leq y_i$ $(i \in \{1,2,3\})$ and $x_i \leq z_i$ $(i \in \{2,3,4\})$. Consequently no coalitionally monotonic core selection exists on any class of games containing these three games; by Lemma 3.1, three is the minimal possible size of such a certificate.*

Proof. In $\mathrm{Core}(v_A)$: adding $y(\{1,2,3\}) \geq 4$, $y(\{1,4\}) \geq 3$ and $y(\{2,4\}) \geq 3$ gives $2y_1 + 2y_2 + y_3 + 2y_4 \geq 10$, i.e. $2\,v_A(N) - y_3 \geq 10$, so $y_3 \leq 2$. In $\mathrm{Core}(v_B)$: adding $z(\{1,3\}) \geq 3$, $z(\{1,4\}) \geq 3$ and $z(\{2,3,4\}) \geq 6$ gives $2z_1 + z_2 + 2z_3 + 2z_4 \geq 12$, so $z_2 \leq 0$. Monotonicity then forces $x_3 \leq y_3 \leq 2$ and $x_2 \leq z_2 \leq 0$, whence $x(\{2,3\}) \leq 2 < 3 = v(\{2,3\})$, contradicting $x \in \mathrm{Core}(v)$. Minimality follows from Lemma 3.1. □

Remark 3.3. *The proof is a rearrangement of an exact Farkas certificate with eleven multipliers (nine core and monotonicity constraints with weight* 1*; the two efficiency identities with weight* 2*), found by linear programming over the family (Section 8; the multiplier table is in the supplement). None of the three games is exact, consistent with the obstructions of Section 7.*

4 The SD-prenucleolus

4.1 Definition

Fix (N, v) and an efficient x $(x(N) = v(N))$. The SD-prenucleolus of Arin–Katsev (Arin and Katsev 2011, 2014) is defined through a recursive *satisfaction* vector, as follows. Let $\mathcal{C} = 2^N \setminus \{\emptyset, N\}$. Maintain a set $H \subseteq \mathcal{C}$ of coalitions whose satisfactions $F(T; x)$ have been fixed. For $S \notin H$ let $H(S) = \bigcup\{T \in H :\ T \subseteq S\}$ (a set of *players*) and, for $i \in H(S)$,

$$f(i,S) \;=\; \min\{F(T;x) :\ T \in H,\ i \in T \subseteq S\}.$$

The current candidate satisfaction of S is

$$F_H(S) = \begin{cases} \dfrac{x(S) - v(S) - \sum_{i \in H(S)} f(i,S)}{|S| - |H(S)|}, & H(S) \neq S, \\[2ex] x(S) - v(S) - \sum_{i \in S} f(i,S) + \max_{i \in S} f(i,S), & H(S) = S. \end{cases}$$

Starting from $H = \emptyset$ (all candidates are then per-capita satisfactions $(x(S) - v(S))/|S|$), repeatedly fix *all* coalitions attaining the minimal current candidate at that value and update; the procedure terminates with the satisfaction vector $F(\cdot\,; x)$. The *SD-prenucleolus* $\mathrm{SD}(N, v)$ is the efficient x whose non-decreasingly sorted satisfaction vector is lexicographically maximal. It is single-valued, continuous, covariant, a core selection on balanced games, strongly aggregate monotonic, and coalitionally monotonic on convex games and on monotonic veto games (Arin and Katsev 2011, 2014, 2016a, 2018).

The solution admits a characterization in the style of Kohlberg (1971) (Arin and Katsev 2011, 2018): an efficient x equals $\mathrm{SD}(N, v)$ if and only if for every threshold $\alpha \in \mathbb{R}$ the lower level set $\{S \in \mathcal{C} : \; F(S; x) \leq \alpha\}$ is empty or balanced. We use this as the certification instrument throughout.

4.2 Implementation and validation

We compute $\mathrm{SD}(N, v)$ by a Maschler-type successive linear-programming scheme (at each stage the candidate satisfactions of unfixed coalitions are affine in x, because fixed satisfactions are constants), with every output certified by (i) an exact evaluation of the recursion above at the candidate point and (ii) the Kohlberg-type criterion, with balancedness of each level set established by an exact linear program producing strictly positive weights. As additional regression checks (not proofs), headline witnesses were probed for lexicographic optimality along hundreds of exact rational perturbation directions.

The implementation reproduces exactly: the worked example of Arin and Katsev (2018); the prenucleolus on two-player games; the serial rule on random monotonic veto games (Theorem 14.3 of Arin and Katsev 2018, using the explicit serial-rule formula of Arin and Feltkamp 2012); the minimal-overlap division on the bankruptcy game $(E; c) = (200; 100, 200, 300)$, namely (100/3, 250/3, 250/3) (Theorem 14.4 of Arin and Katsev 2018), which differs from the Talmud/nucleolus division $(50, 75, 75)$; strict strong aggregate monotonicity on 150 random games; and coalitional monotonicity along 8 random convex bumps. On the five-player benchmark game of Arin–Iñarra reported in Iñarra et al. (2020) our classical nucleolus solver reproduces the published value $(16/9, 16/9, 16/9, 4/3, 4/3)$ exactly.

5 The boundary of the SD-prenucleolus

5.1 Coalitional monotonicity fails on exact games

Theorem 5.1. *The SD-prenucleolus is not coalitionally monotonic on the class of exact games, already for $n = 4$. Witness: let v be*

$$v(\{2,3\}) = 1,\;\; v(\{1,2,3\}) = 1,\;\; v(\{2,4\}) = 13,\;\; v(\{1,2,4\}) = 15,$$

$$v(\{1,3,4\}) = 14,\;\; v(\{2,3,4\}) = 13,\;\; v(N) = 30,$$

all other coalitions 0*, and let $v' = v + 1 \cdot e_{\{1,3,4\}}$. Both games are exact (hence totally balanced; neither is convex). Then*

$$\mathrm{SD}(v) = \left(\tfrac{53}{12},\;\tfrac{19}{4},\;\tfrac{15}{4},\;\tfrac{205}{12}\right), \qquad \mathrm{SD}(v') = \left(\tfrac{35}{8},\;\tfrac{33}{8},\;\tfrac{33}{8},\;\tfrac{139}{8}\right),$$

so player $1 \in \{1, 3, 4\}$ loses $1/24$ when the worth of a coalition containing her rises.

Proof (computer-assisted). Exactness of each game is certified by $2^n - 2$ exact linear programs exhibiting, for every coalition S, a core element with $x(S) = v(S)$. That the displayed points are the SD-prenucleoli follows from the Kohlberg-type criterion of Section 4: the full satisfaction level structure of both points and strictly positive balancing weights for the lower level sets are listed in Appendix A and can be checked by hand. The payoff comparison is exact rational arithmetic. □

The witness was found by search and greedily minimized (total mass 87, seven nonzero values); both games are moreover superadditive, so the failure holds on exact (hence totally balanced) superadditive games. Since Arin–Katsev prove monotonicity on convex games, the boundary along convex $\subsetneq$ exact $\subsetneq$ totally balanced $\subsetneq$ balanced is *convexity itself.*

Corollary 5.2. *Exactness does not imply SD-relevance, the domain hypothesis of the general monotonicity theorem of Arin and Katsev (2018).*[2]

5.2 The SD-prekernel is multi-valued on exact games

Arin and Katsev (2016b, 2018) define the SD-prekernel through bilateral complaints $s_{ij}(x) = \min\{F(S;x) : i \in S,\ j \notin S\}$: it is the set of efficient x with $s_{ij}(x) = s_{ji}(x)$ for all $i \neq j$. It contains the SD-prenucleolus, and on convex games it equals it (a singleton).

Theorem 5.3. *The SD-prekernel can be multi-valued—indeed positive-dimensional—on exact games. For the exact, totally balanced, non-convex game*

$$v(\{1\}) = 5,\ v(\{2\}) = 6,\ v(\{3\}) = 2,\ v(\{4\}) = 5,$$
$$v(\{1,2\}) = 20,\ v(\{1,3\}) = 10,\ v(\{2,3\}) = 8,\ v(\{1,4\}) = 12,\ v(\{2,4\}) = 14,\ v(\{3,4\}) = 12,$$
$$v(\{1,2,3\}) = 22,\ v(\{1,2,4\}) = 25,\ v(\{1,3,4\}) = 17,\ v(\{2,3,4\}) = 18,\quad v(N) = 32,$$

the SD-prekernel contains the segment $\mathrm{SD}(v) + t\,(1,-1,-1,1)$, $t \in [-1/2, 1/2]$, *where* $\mathrm{SD}(v) = (19/2,\ 21/2,\ 9/2,\ 15/2)$*; every point of the segment lies in the core and equalizes all six bilateral complaints exactly.*

Proof. Write $x^t = \mathrm{SD}(v) + t\,(1,-1,-1,1)$ for $t \in [-1/2, 1/2]$. Core membership of x^t reduces to that of the endpoints $(9, 11, 5, 7)$ and $(10, 10, 4, 8)$, which is a direct check against the fifteen constraints. Evaluating the recursion of Section 4 at x^t gives, for every $t \in [-1/2, 1/2]$,

$$F_{\{1,2\}} = F_{\{3,4\}} = 0, \qquad F_{\{1,3\}} = F_{\{2,4\}} = 2,$$

$$F_{\{3\}} = F_{\{1,2,3\}} = \tfrac{5}{2} - t, \qquad F_{\{4\}} = F_{\{1,4\}} = F_{\{1,2,4\}} = \tfrac{5}{2} + t,$$
$$F_{\{2\}} = F_{\{2,3\}} = F_{\{2,3,4\}} = \tfrac{9}{2} - t, \qquad F_{\{1\}} = F_{\{1,3,4\}} = \tfrac{9}{2} + t$$

The displayed functions are proved by running the recursion symbolically, stage by stage, on each open half-interval; the fixing order is constant on each. For $t \in (-\frac{1}{2}, 0)$ the stages are

$$\{12,34\};\ \{13,24\};\ \{4,14,124\};\ \{3,123\};\ \{1,134\};\ \{2,23,234\}$$

[2] We refer to Arin and Katsev (2018) for the definition of SD-relevant games; here we use only the fact that their theorem establishes coalitional monotonicity of the SD-prenucleolus on that class, which by Theorem 5.1 therefore cannot contain both witness games.

at levels 0, 2, $\frac{5}{2}+t$, $\frac{5}{2}-t$, $\frac{9}{2}+t$, $\frac{9}{2}-t$ respectively, and for $t \in (0, \frac{1}{2})$ they are

$$\{12, 34\};\ \{13, 24\};\ \{3, 123\};\ \{4, 14, 124\};\ \{2, 23, 234\};\ \{1, 134\}$$

at levels 0, 2, $\frac{5}{2}-t$, $\frac{5}{2}+t$, $\frac{9}{2}-t$, $\frac{9}{2}+t$. At each stage, direct substitution of x^t into the two-case formula of Section 4, using the already-fixed values, yields exactly the displayed candidates and confirms that the stated minimizers are fixed next (the required inequalities, e.g. $2 \le \frac{5}{2} \mp t$ and $\frac{5}{2}+t < \frac{5}{2}-t$ for $t<0$, hold strictly on the open half-intervals). The boundary values $t \in \{-\frac{1}{2}, 0, \frac{1}{2}\}$ are verified separately by direct exact computation (supplement), where some stages merge but the resulting satisfaction values still obey the displayed formulas. Reading off the bilateral complaints $s_{ij} = \min\{F_S :\ i \in S \not\ni j\}$: for $|t| \le \frac{1}{2}$, $s_{12} = s_{21} = 2$ (attained at $\{1,3\}$ resp. $\{2,4\}$), $s_{34} = s_{43} = 2$, and all other ordered pairs have $s_{ij} = 0$ (attained at $\{1,2\}$ or $\{3,4\}$). All six equalities $s_{ij} = s_{ji}$ therefore hold along the segment. That the midpoint is $\mathrm{SD}(v)$ is certified by the Kohlberg-type criterion (Appendix A). □

Thus the uniqueness theorem for convex games is tight, with the same boundary as Theorem 5.1. (Perturbed control points off the segment fail the equalities, confirming the checker discriminates.)

5.3 Population monotonicity fails on convex games

Theorem 5.4. *The SD-prenucleolus is not population monotonic on convex games, already at $n = 3$, which is minimal: for $n \le 2$ every core selection is population monotonic, since individual rationality gives $\theta_i(v|_{\{i\}}) = v(\{i\}) \le \theta_i(v|_T)$ for any $T \ni i$. For the convex, exact game $v(\{1,2\}) = v(\{2,3\}) = 1$, $v(N) = 2$ (all else 0):*

$$\mathrm{SD}\big(v|_{\{2,3\}}\big) = \big(\tfrac{1}{2}, \tfrac{1}{2}\big), \qquad \mathrm{SD}(v) = \big(\tfrac{1}{3},\ \tfrac{4}{3},\ \tfrac{1}{3}\big),$$

so player 3 falls from 1/2 to 1/3 when player 1 joins.

Proof. Convexity is immediate. The two-player subgame on $\{2,3\}$ has $v(\{2\}) = v(\{3\}) = 0$, $v(\{2,3\}) = 1$; the SD-prenucleolus coincides with the prenucleolus on two-player games, giving the equal split $(1/2, 1/2)$.

For the full game we verify $x = (1/3, 4/3, 1/3)$ by the Kohlberg-type criterion. The recursion finishes in two stages (the five coalitions at the minimal level are fixed first, then $\{2\}$): the per-capita satisfactions are $F(\{1\}) = F(\{3\}) = 1/3$; $F(\{1,2\}) = F(\{2,3\}) = (x(S)-1)/2 = 1/3$; $F(\{1,3\}) = x(\{1,3\})/2 = 1/3$; $F(\{2\}) = 4/3$. The lower level sets are: for $\alpha \in [1/3, 4/3)$, $\mathcal{B}_1 = \{\{1\}, \{3\}, \{1,2\}, \{2,3\}, \{1,3\}\}$, balanced with strictly positive weights $\lambda_{\{1,2\}} = \lambda_{\{2,3\}} = \frac{1}{2}$, $\lambda_{\{1,3\}} = \frac{1}{3}$, $\lambda_{\{1\}} = \lambda_{\{3\}} = \frac{1}{6}$ (player 1: $\frac{1}{2}+\frac{1}{3}+\frac{1}{6} = 1$; player 2: $\frac{1}{2}+\frac{1}{2} = 1$; player 3: $\frac{1}{3}+\frac{1}{2}+\frac{1}{6} = 1$); for $\alpha \ge 4/3$, the set of all proper coalitions, which is balanced. Hence every lower level set is empty or balanced and $x = \mathrm{SD}(v)$. Finally $1/2 > 1/3$. □

In exploratory float-screened sampling of convex games (integer-valued supermodular games generated as nonnegative integer combinations of unanimity games), violations appeared in roughly half of the instances at $n = 4$ and more at $n = 5$; we report these frequencies as indicative only, since the exploratory counting runs were not preserved as seeded scripts. The phenomenon is in any case common under such sampling, not exceptional. The SD-prenucleolus therefore separates the two monotonicities on its home class: it is coalitionally monotonic there (Arin and Katsev 2016a) but not population monotonic—in contrast with the egalitarian solution of Dutta and Ray (1989), which is population monotonic on convex games.

6 Market games: pairwise monotonicity

An *assignment game* (Núñez and Rafels 2003) is generated by a nonnegative matrix $A = (a_{ij})$ over two disjoint sides M (rows) and M' (columns): the player set is $M \cup M'$ and $v(S)$ is the maximum weight of a matching between $S \cap M$ and $S \cap M'$ with weights a_{ij}. A *permutation game* (Tijs et al. 1984) is generated by a square matrix $K = (k_{ij})$ over a single player set: $v(S) = \max_\pi \sum_{i \in S} k_{i\pi(i)}$, the maximum over permutations π of S. Assignment games embed into permutation games (Curiel and Tijs 1986); both classes are totally balanced.

Pairwise monotonicity (Núñez and Rafels 2003) is the natural monotonicity axiom for such games: if one entry of the matrix rises, neither of the two agents named by that entry may lose. The buyers-optimal, sellers-optimal and τ-value rules satisfy it on assignment games (Núñez and Rafels 2003; van den Brink et al. 2021); Miquel (2009) constructs a bespoke pairwise-monotonic core selection for permutation games.[3] For the nucleolus on assignment games, the 2008 OTKA project report of Solymosi, Forgó and Pintér (Solymosi et al. 2008) announces: "We proved that the nucleolus is pairwise-monotonic on assignment games"; the project's 2012 final report attributes the result to an unpublished manuscript, "Pairwise monotonicity of the nucleolus in assignment games" by T. Solymosi, M. Brugueras and T. E. S. Raghavan (2012), with a simplified proof in a revised version.[4] We have not been able to access either manuscript; our computations below independently support the statement.

Theorem 6.1. *The nucleolus is not pairwise monotonic on permutation games. Let K be the value matrix (rows: owners; columns: recipients)*

$$K = \begin{pmatrix} 0 & 7 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 9 & 1 & 0 \end{pmatrix},$$

and raise K_{43} from 1 to 2 (which raises only $v(\{3,4\})$ and $v(\{1,3,4\})$, each by one, and fixes $v(N) = 11$). The nucleolus moves from $(\frac{1}{2}, \frac{33}{4}, \frac{1}{2}, \frac{7}{4})$ to $(\frac{2}{3}, \frac{23}{3}, \frac{1}{3}, \frac{7}{3})$: player 3, a member of both improved coalitions, falls from $1/2$ to $1/3$. Both games are totally balanced and both nucleoli lie in the core. Moreover the SD-prenucleolus fails on the same instance, falling from $1/2$ to $1/4$ for the same player.

Proof (computer-assisted). Both nucleolus claims are certified by Kohlberg's criterion in exact arithmetic; Appendix A lists the excess level structure and balancing weights at the top level for both games, and the machine-readable supplement contains the full level data. The SD claims are certified as in Section 4.2. □

The witness is greedily minimized (integer matrix, total mass 20); a float-screened $n = 3$ sweep of 36,000 bumps found no candidate violation, so $n = 4$ is the smallest size at which we found the phenomenon. In this instance the improving pair's *joint* payoff rises ($9/4 \to 8/3$ for the nucleolus), so the witness does not violate Sasaki's weak (joint) pairwise monotonicity (Sasaki 1995); the bump's surplus is redistributed *within* the improving pair.

[3] We could not access the body of Miquel (2009); its abstract states a positive result for a merged-τ-value rule and makes no claim about the nucleolus. Whether an explicit nucleolus counterexample appears in its introduction is a caveat for the novelty (not the correctness) of Theorem 6.1.

[4] OTKA final report, `https://real.mtak.hu/12261/1/72856_ZJ1.pdf`.

On assignment games, by contrast, an extensive search—float-screened random and tie-targeted 3×3, 4×3, 4×4 markets and hill-climbing on the violation margin, together with one fully exact component, an exhaustive Kohlberg-certified sweep of all 2×2 markets with entries in $\{0, \ldots, 4\}$ (2,500 bumps, zero violations, zero certification failures)—found no violation for either solution. For the nucleolus this is consistent with (Solymosi et al. 2008); for the SD-prenucleolus we know of no prior statement:

Conjecture 6.2. *The SD-prenucleolus is pairwise monotonic on assignment games.*

Conjecture 6.3. *The nucleolus of minimum-cost spanning tree games is cost monotonic in the sense of Dutta and Kar (2004) (raising one edge's cost never lowers an incident player's cost share).*

Conjecture 6.3 is supported by 1,450 float-screened tests (with exact certification awaiting any candidate violation; none arose) at $n \in \{4, 5, 6\}$ including large bumps in the core-moving regime of Tan et al. (2025); the nucleolus is known to be monotonic on the fixed-tree subclass (Maschler et al. 2010), and Dutta–Kar's violating rule is Bird's, not the nucleolus. Theorem 6.1 shows any proof of the assignment-game statements must use two-sidedness: the property fails one class up.

7 Do exact games escape the impossibility?

The certificate games of Theorem 3.2, like those of Housman and Clark (1998), are not exact. This section first records that for $n \leq 4$ the existence question on exact games is settled affirmatively by known results (Proposition 7.29), then shows that the pinning and two-arm mechanisms underlying the known impossibility proofs are unavailable on exact games of any size, delimits the techniques by an explicit counterexample to a natural strengthening, and reduces the existence question to finite-family feasibility.

7.1 The pinning mechanism is unavailable

The Housman–Clark proof pins players via bumps whose cores collapse to singletons.

Lemma 7.1. *An exact game whose core is a singleton is additive. Consequently no exact game family can pin a player through a singleton core except trivially.*

Proof. If $\mathrm{Core}(v) = \{x\}$, exactness gives, for each S, a core element attaining $x(S) = v(S)$; but the core element is unique, so $x(S) = v(S)$ for every S, i.e. v is additive with $v(S) = \sum_{i \in S} x_i$. □

More generally, pinning arguments transmit *face bounds* $\max_{x \in \mathrm{Core}} x(T)$ along monotonicity chains. On exact games these bounds cannot contract under proper bumps:

Theorem 7.2 (Pin invariance under proper bumps)**.** *Let all games below be exact and let all bumps be proper. Fix a nonempty group $T \subseteq N$. Along any monotonicity chain that constrains T—i.e. any sequence of proper bumps at coalitions $S \supseteq T$—the face bound of Lemma 2.2 satisfies $\max_{x \in \mathrm{Core}} x(T) = v(N) - v(N \setminus T)$ in every game of the chain, with both $v(N)$ and $v(N \setminus T)$ unaltered by every bump in the chain. In particular, since face bounds over any partition sum to at least $v(N)$ (Lemma 2.2), no contradiction with efficiency can be derived from partition-aggregated face bounds, at any chain depth and for groups of any arity; we do not claim the analogue for arbitrary balanced collections of groups. (A bump at $S = N$ raises every face bound by d and is likewise harmless.)*

Proof. A proper bump at S constrains T only if $T \subseteq S$; since $T \cap (N \setminus T) = \emptyset$ and $S \neq N$, no such S equals $N \setminus T$ or N, so neither $v(N \setminus T)$ nor $v(N)$—and with them the face bound, by Lemma 2.2 applied in each (exact) game—changes along the chain. □

7.2 Two-arm families: positive results and their limits

Lemma 7.3 (Overlapping arms)**.** *Let b be exact and let $t_A = b + d_A e_A$, $t_B = b + d_B e_B$ be exact, where $A \neq B$ are nonempty proper coalitions with $A \cap B \neq \emptyset$. Then $M = b + d_A e_A + d_B e_B$ is balanced, and any $x^* \in \mathrm{Core}(t_A)$ maximizing $x(B)$ lies in $\mathrm{Core}(M) \subseteq \mathrm{Core}(b) \cap \mathrm{Core}(t_A) \cap \mathrm{Core}(t_B)$; assigning x^* to all three games shows the family $\{b, t_A, t_B\}$ is feasible.*

Proof. By exactness of t_A at $N \setminus B$ and $A \not\subseteq N \setminus B$ (as $A \cap B \neq \emptyset$),

$$\max_{x \in \mathrm{Core}(t_A)} x(B) = t_A(N) - t_A(N \setminus B) = b(N) - b(N \setminus B) = \max_{x \in \mathrm{Core}(b)} x(B) \;\geq\; b(B) + d_B,$$

the last inequality because $\mathrm{Core}(t_B) \neq \emptyset$ forces some core element of b with $x(B) \geq b(B) + d_B$. The maximizer x^* then satisfies every constraint of M. □

Lemma 7.4 (Singleton arms are transparent)**.** *Let b and $t_i = b + d e_{\{i\}}$ be balanced. For every $x \in \mathrm{Core}(b)$ there is $y \in \mathrm{Core}(t_i)$ with $y_i \geq x_i$. Consequently, in a three-game star $\{b, t_A, t_B\}$, if one arm is a singleton then the family is feasible (without any exactness assumption).*

Proof. Put $u = \max\{z_i : z \in \mathrm{Core}(b)\}$. Balancedness of t_i means that some core point of b satisfies $z_i \geq b(\{i\}) + d$, so $u \geq b(\{i\}) + d$. A maximizer y of z_i over $\mathrm{Core}(b)$ therefore belongs to $\mathrm{Core}(t_i)$ and satisfies $y_i = u \geq x_i$ for every $x \in \mathrm{Core}(b)$.

For the consequence, suppose $A = \{i\}$. Choose any $x \in \mathrm{Core}(t_B) \subseteq \mathrm{Core}(b)$ and assign it to both b and t_B. The first statement supplies $y \in \mathrm{Core}(t_A)$ with $y_i \geq x_i$, which completes the family witness. □

Corollary 7.5 (complete reduction for two-arm stars)**.** *Let b, t_A, t_B be exact single-bump games. The star family $\{b, t_A, t_B\}$ is feasible whenever $A \cap B \neq \emptyset$ or $\min\{|A|, |B|\} = 1$. Hence every such family on at most three players is feasible. Any case not covered by these two local lemmas must have*

$$A \cap B = \emptyset, \qquad |A| \geq 2, \qquad |B| \geq 2,$$

beginning with a two-pair partition at $n = 4$. Proposition 7.29 settles every four-player family globally, so the open two-arm frontier starts at $n = 5$.

Proof. Use Lemma 7.3 in the overlapping case and Lemma 7.4 in the singleton case. Two disjoint nonempty coalitions on at most three players cannot both have size at least two. The final statement uses Proposition 7.29. □

Proposition 7.6 (both bump rows are necessary)**.** *Consider a three-game star $\{b, t_A, t_B\}$ of balanced games, with $t_A = b + d_A e_A$ and $t_B = b + d_B e_B$. In any Farkas certificate of infeasibility for its family LP, the multipliers on the A-core constraint of t_A and the B-core constraint of t_B are both strictly positive. Thus the three-game question reduces to Farkas rays simultaneously sensitive to the two changed worth rows.*

Proof. Suppose, for example, that the multiplier on the B-constraint of t_B is zero. Replace t_B by b, equivalently set $d_B = 0$. The Farkas right-hand side is unchanged, since this is the only row whose worth changes. But the resulting family $\{b, t_A, b\}$ is feasible: choose any $x \in \mathrm{Core}(t_A) \subseteq \mathrm{Core}(b)$ and assign x to all three games. The same purported certificate would therefore prove a feasible system infeasible, a contradiction. The argument for the A-row is symmetric. □

Lemma 7.7 (positive-circuit formulation)**.** *Fix the player set and the two labeled arms. Let H be the inequality matrix of the three-game family LP, choose a reference player r, and form D by replacing, within each game block, the stationarity equation for player $i \neq r$ by its difference from the equation for r. Then the projection of the Farkas cone onto its nonnegative inequality multipliers is*

$$\{\lambda \geq 0 : D\lambda = 0\}.$$

Its extreme rays are exactly the support-minimal positive circuits of D. Consequently, if no positive circuit contains both changed-worth core rows, then every balanced three-game star with those arm labels is feasible.

Proof. The only equality rows in the family LP are the three efficiency equations. For a fixed game, a free efficiency multiplier completes stationarity if and only if the inequality-row combination has the same coefficient on every player. Subtracting the reference-player coefficient gives exactly $D\lambda = 0$, and then the efficiency multiplier is uniquely recovered. Thus the displayed cone is linearly isomorphic to the full Farkas cone. Extreme rays of a nonnegative kernel cone are precisely its nonzero support-minimal nonnegative dependencies, i.e. its positive circuits.

If the family were infeasible, a Farkas certificate with negative right-hand side would decompose into extreme rays, at least one of which would still have negative right-hand side. Proposition 7.6 forces that ray to use both changed-worth rows. The corresponding positive circuit would therefore contain both distinguished columns, a contradiction. □

Two exact support reductions further sharpen the target. A both-sensitive positive circuit must use a monotonicity column from each arm: if, say, no A-edge column were used, stationarity in the t_A block would make its supported core columns alone a nonzero positive dependency, contradicting support minimality. Also, it cannot contain both core columns T and $N \setminus T$ in one game block, because those two columns sum to zero and already form a positive circuit.

This formulation permits a direct search that is less likely to miss a ray with a small normal cone than random optimization over a normalized section. The exactified overlapping-arm positive control is recovered as a nullity-one positive circuit. A sparse-support search also corrects the initial lesson from 600,000 targeted bases, none of which found a disjoint circuit: both-sensitive circuits do exist for disjoint arms at $n = 5$. For example, with $A = \{1, 2, 3\}$ and $B = \{4, 5\}$ there is a nullity-one unit circuit on the base core row $\{3, 5\}$; the t_A rows $A, \{1, 4, 5\}, \{2, 4, 5\}$; the t_B rows $\{1, 3\}, \{2, 3\}, \{1, 2, 4\}, B$; and the monotonicity rows for player 3 on the A-arm and player 5 on the B-arm. Its efficiency weights are $(0, 2, 2)$.

Thus the no-common-circuit route is false. The relevant circuit inequality is nevertheless forced by exactness:

$$\begin{aligned} 4b(N) \;\geq\; & b(\{3,5\}) + b(A) + d_A + b(\{1,4,5\}) + b(\{2,4,5\}) \\ & + b(\{1,3\}) + b(\{2,3\}) + b(\{1,2,4\}) + b(B) + d_B. \end{aligned}$$

Indeed, exactness of both bumped games at the single coalition $\{1\}$ gives

$$t_A(A) + t_A(\{1,4,5\}) \le t_A(N) + t_A(\{1\}),$$
$$t_B(\{1,3\}) + t_B(\{2,3\}) + t_B(\{1,2,4\}) + t_B(\{3,5\}) + t_B(B) + t_B(\{2,4,5\}) \le 3t_B(N) - t_B(\{1\}).$$

The first collection covers player 1 twice and every other player once; the second covers player 1 twice and every other player three times. Because the bump coalitions are A and B, all other displayed worths are base worths. Adding cancels $b(\{1\})$ and proves the circuit inequality. In contrast, dropping endpoint exactness makes the circuit inequality violable and gives a rational balanced disjoint-arm family with Farkas right-hand side -1. The central question is therefore whether *every* both-sensitive disjoint circuit inequality is implied by endpoint exactness, not whether such circuits are absent.

The first circuit generates six distinct supports under permutations within A and B. Blocking that entire orbit exposes a second ten-row unit-circuit type (with a twelve-support orbit): base row $\{1,5\}$; t_A rows $A, \{2,4,5\}, \{3,4,5\}$; t_B rows $\{1,2\}, \{2,3\}, \{1,3,4\}, B$; and monotonicity rows for players 1 and 5. This second type is already neutralized by balancedness. In t_A, the rows

$$\tfrac{1}{2}\{1,2\},\ A,\ \tfrac{1}{2}\{1,3,4\},\ \tfrac{1}{2}\{1,5\},\ \{2,4,5\},\ \{3,4,5\}$$

form a balanced collection of total weight $5/2$; in t_B, the rows

$$\tfrac{1}{2}\{1,2\},\ \{2,3\},\ \tfrac{1}{2}\{1,3,4\},\ \tfrac{1}{2}\{1,5\},\ B$$

form one of total weight $3/2$. Summing their balancedness inequalities is exactly the circuit inequality.

Exactness first supplies a small extension of the usual balancedness inequality.

Lemma 7.8 (two-level exactness inequality)**.** *Let v be exact, fix a proper coalition U, and let $w_S \ge 0$ satisfy*

$$\sum_{S \ni i} w_S = \alpha + \delta \mathbf{1}_{\{i \in U\}} \qquad (i \in N)$$

for constants α, δ (with the displayed coverage nonnegative). Then

$$\sum_S w_S v(S) \le \alpha v(N) + \delta v(U).$$

Proof. Choose $x \in \mathrm{Core}(v)$ with $x(U) = v(U)$. Summing the core inequalities with weights w_S gives

$$\sum_S w_S v(S) \le \sum_i (\alpha + \delta \mathbf{1}_{\{i \in U\}}) x_i = \alpha v(N) + \delta v(U).$$

□

Balancedness is the special case $\delta = 0$. The mechanism behind the recoloring certificate has a second useful formulation that also does not depend on five players. Write $\lambda_{g,S}$ for the core-row multipliers of a positive circuit, where $g = 0, 1, 2$ denotes b, t_A, t_B, and write a_g for its efficiency weights.

Lemma 7.9 (balanced recoloring criterion)**.** *Put $q_S = \sum_g \lambda_{g,S}$. Then q is a balanced collection of total weight $\sum_g a_g$. Suppose it can be decomposed as $q_S = \sum_g \mu_{g,S}$ so that each μ_g is balanced, and*

$$\mu_{1,A} \geq \lambda_{1,A}, \qquad \mu_{2,B} \geq \lambda_{2,B}.$$

Then the circuit inequality is valid for every balanced base-plus-two-bump triple with $d_A, d_B \geq 0$.

Proof. Sum the circuit stationarity equations over the three game blocks. Every monotonicity row cancels between its source and target, leaving

$$\sum_{S \ni i} q_S = \sum_g a_g \qquad (i \in N),$$

which proves the first claim. Let β_g be the total weight of μ_g. Balancedness of the three games and their common grand worth give

$$\begin{aligned}
\Big(\sum_g a_g\Big) b(N) &= \sum_g \beta_g v_g(N) \\
&\geq \sum_{g,S} \mu_{g,S} v_g(S) \\
&= \sum_S q_S b(S) + \mu_{1,A} d_A + \mu_{2,B} d_B \\
&\geq \sum_{g,S} \lambda_{g,S} v_g(S).
\end{aligned}$$

This is precisely nonnegativity of the circuit's Farkas right-hand side. □

There is a one-witness extension that isolates the role of exactness while retaining the coloring interpretation.

Lemma 7.10 (single-witness recoloring criterion)**.** *Fix a proper coalition U. Suppose there are weights $\mu_{g,S} \geq 0$ and scalars α_g, δ_g such that*

$$\sum_{S \ni i} \mu_{g,S} = \alpha_g + \delta_g \mathbf{1}_{\{i \in U\}} \qquad (g = 0, 1, 2;\ i \in N).$$

Put $\Delta = \sum_g \delta_g$ and suppose

$$\begin{gathered}
\sum_g \mu_{g,S} = q_S + \Delta \mathbf{1}_{\{S = U\}}, \\
\sum_g \alpha_g = \sum_g a_g, \\
\mu_{1,A} - \delta_1 \mathbf{1}_{\{U = A\}} \geq \lambda_{1,A}, \qquad \mu_{2,B} - \delta_2 \mathbf{1}_{\{U = B\}} \geq \lambda_{2,B}.
\end{gathered}$$

Then the circuit inequality is valid for every exact endpoint triple.

Proof. Apply Lemma 7.8 to each color and sum:

$$\sum_{g,S} \mu_{g,S} v_g(S) \leq \Big(\sum_g \alpha_g\Big) b(N) + \sum_g \delta_g v_g(U).$$

The aggregate identity makes the base-worth part of the left side $\sum_S q_S b(S) + \Delta b(U)$, while the right side contains the same $\Delta b(U)$. These terms cancel. The only remaining nonbase terms on the right are $\delta_1 d_A$ if $U = A$ and $\delta_2 d_B$ if $U = B$; the adjusted retention inequalities therefore leave at least $\lambda_{1,A} d_A + \lambda_{2,B} d_B$ on the left. The efficiency identity then gives exactly the circuit bound. □

Lemma 7.9 is the special case $\delta_g = 0$. The advantage of the extension is that a failed balanced coloring can be repaired by one coalition-tight witness, provided the aggregate row inserted at U is exactly offset by the three coverage defects.

An especially useful specialization removes the base color and the aggregate insertion altogether.

Corollary 7.11 (two-color transfer criterion)**.** *Fix a proper coalition U. Suppose there are weights $0 \le w_S \le q_S$ and scalars α, δ such that*

$$\sum_{S \ni i} w_S = \alpha + \delta \mathbf{1}_{\{i \in U\}} \qquad (i \in N),$$

and

$$w_A - \delta \mathbf{1}_{\{U=A\}} \ge \lambda_{1,A},$$
$$q_B - w_B + \delta \mathbf{1}_{\{U=B\}} \ge \lambda_{2,B}.$$

Then the circuit inequality is valid for every exact endpoint triple.

Proof. Put $a = \sum_g a_g$. In Lemma 7.10, set

$$\mu_{0,S} = 0, \qquad \mu_{1,S} = w_S, \qquad \mu_{2,S} = q_S - w_S,$$

and take $(\alpha_0, \delta_0) = (0, 0)$, $(\alpha_1, \delta_1) = (\alpha, \delta)$, and $(\alpha_2, \delta_2) = (a - \alpha, -\delta)$. Since q is balanced of total weight a, the second colored collection has coverage $(a - \alpha)\mathbf{1} - \delta \mathbf{1}_U$. Thus all hypotheses of that lemma hold with $\Delta = 0$. □

The corollary converts the repair problem into a box-constrained degree problem: find a subcollection w of the aggregate balanced collection whose degree vector lies in $\mathrm{span}\{\mathbf{1}, \mathbf{1}_U\}$ while retaining the two distinguished rows. This one-color formulation is the natural starting point for a cut or zonotope proof.

Lemma 7.12 (base-color elimination)**.** *The recoloring capacity equals the largest ρ for which there are $0 \le w_S \le q_S$ and α satisfying*

$$\sum_{S \ni i} w_S = \alpha \quad (i \in N), \qquad w_A \ge \rho \lambda_{1,A}, \qquad q_B - w_B \ge \rho \lambda_{2,B}.$$

Thus the unbumped color is redundant in the capacity problem.

Proof. The displayed system gives the coloring $(\mu_0, \mu_1, \mu_2) = (0, w, q - w)$. Conversely, from any balanced coloring (μ_0, μ_1, μ_2) put $w = \mu_1 + \theta \mu_0$ for any $0 \le \theta \le 1$. Then w and $q - w$ are balanced, while

$$w_A \ge \mu_{1,A}, \qquad q_B - w_B = \mu_{2,B} + (1 - \theta)\mu_{0,B} \ge \mu_{2,B}.$$

Hence no retained mass is lost. □

For a proper U, let τ_U be the largest common retained fraction when the two-color transfer system is used: replace the two right sides in Corollary 7.11 by $\tau_U\lambda_{1,A}$ and $\tau_U\lambda_{2,B}$. Write $z(S)=\sum_{i\in S} z_i$ and $[x]_+=\max\{x,0\}$.

Proposition 7.13 (box-capacity separation)**.** *Put*

$$\Phi(r_A,r_B,z)=r_Bq_B+\sum_S q_S\big[z(S)+r_A\mathbf{1}_{\{S=A\}}-r_B\mathbf{1}_{\{S=B\}}\big]_+.$$

Then

$$\rho=\min\ \Phi(r_A,r_B,z),$$

where the minimum is over

$$r_A,r_B\ge 0,\qquad r_A\lambda_{1,A}+r_B\lambda_{2,B}=1,\qquad z(N)=0.$$

Moreover,

$$\tau_U=\min\ \Phi(r_A,r_B,z)$$

over the same set with the additional equation

$$z(U)+r_A\mathbf{1}_{\{U=A\}}-r_B\mathbf{1}_{\{U=B\}}=0.$$

In particular, $\tau_U\ge\rho$.

Proof. Use the reduced primal of Lemma 7.12, adding the free two-level defect δ for the fixed-U problem. Let $-r_A,-r_B$ be the dual multipliers of the two retention inequalities and let z_i be the multiplier of player i's coverage equation. Stationarity of the free variables α,δ, and τ_U gives, respectively,

$$z(N)=0,\qquad z(U)+r_A\mathbf{1}_{\{U=A\}}-r_B\mathbf{1}_{\{U=B\}}=0,\qquad r_A\lambda_{1,A}+r_B\lambda_{2,B}=1.$$

For each box variable $0\le w_S\le q_S$, eliminating its two bound multipliers contributes $q_S[z(S)+r_A\mathbf{1}_{\{S=A\}}-r_B\mathbf{1}_{\{S=B\}}]_+$ to the dual objective. The constant q_B in the second retention inequality contributes r_Bq_B. Strong LP duality gives the formula for τ_U; omitting δ and its stationarity equation gives the formula for ρ. □

Define the support-nullspace

$$K(q)=\{h\in\mathbb{R}^N : h(N)=0,\ h(S)=0 \text{ whenever } q_S>0\}.$$

Lemma 7.14 (support-nullspace degeneracy)**.** *If some $h\in K(q)$ has $h(U)\ne 0$, then $\tau_U=\rho$. Equivalently, a fixed witness can improve capacity only if*

$$\mathbf{1}_U\in\operatorname{span}\big(\{\mathbf{1}_N\}\cup\{\mathbf{1}_S : q_S>0\}\big).$$

Proof. Let (r_A,r_B,z) minimize the ordinary formula in Proposition 7.13. For $h\in K(q)$, replacing z by $z+th$ preserves $z(N)=0$ and leaves every positive-weight summand in Φ unchanged. If $h(U)\ne 0$, choose t so that

$$(z+th)(U)+r_A\mathbf{1}_{\{U=A\}}-r_B\mathbf{1}_{\{U=B\}}=0.$$

This gives a feasible fixed-U dual with value ρ. Since $\tau_U\ge\rho$, equality follows. The equivalent span condition is the finite-dimensional orthogonal-complement theorem. □

Call players i, j *support twins* if $\mathbf{1}_{\{i\in S\}} = \mathbf{1}_{\{j\in S\}}$ for every S with $q_S > 0$.

Lemma 7.15 (twin quotient)**.** *Suppose $K(q)$ is spanned by the differences $e_i - e_j$ of support twins. If $z \perp K(q)$, then every coordinate level set of z is support-rigid.*

Proof. Orthogonality gives $z_i = z_j$ for every pair of support twins, so every level set is a union of whole twin classes. Every generator $e_i - e_j$ has sum zero on such a union; hence $h(U) = 0$ for every $h \in K(q)$. Apply the equivalent span condition in Lemma 7.14. □

Because Φ is constant along $K(q)$, orthogonal projection onto $K(q)^\perp$ sends every ordinary minimizer to an ordinary minimizer. Call the result its *support-orthogonal representative*. This removes exactly the gauge freedom in Lemma 7.14.

Proposition 7.13 now isolates a canonical conjecture. For an optimal support-orthogonal potential z, put

$$P^+ = \operatorname{argmax}_{i\in N} z_i, \qquad P^- = \operatorname{argmin}_{i\in N} z_i.$$

Conjecture 7.16 (two-extreme-set repair)**.** *For every both-sensitive positive circuit with disjoint arms and $\rho < 1$, there is an optimal support-orthogonal minimizer with nonconstant z such that*

$$\min\{\tau_{P^+}, \tau_{P^-}\} \geq 1.$$

The value one in this conjecture is not an arbitrary normalization. The box constraints give the following immediate ceiling, which becomes sharp on the entire deficient frontier found below.

Lemma 7.17 (exclusive-arm ceiling)**.** *For every proper U,*

$$U \neq A \implies \tau_U \leq \frac{q_A}{\lambda_{1,A}}, \qquad U \neq B \implies \tau_U \leq \frac{q_B}{\lambda_{2,B}}.$$

In particular, if $q_A = \lambda_{1,A}$ and $U \neq A$ (or symmetrically for B), then any successful full-bump repair has $\tau_U = 1$ exactly.

Proof. If $U \neq A$, the first retention inequality in the two-color transfer system is $w_A \geq \tau_U \lambda_{1,A}$. Since $w_A \leq q_A$, the first bound follows. If $U \neq B$, the second retention inequality is $q_B - w_B \geq \tau_U \lambda_{2,B}$; use $w_B \geq 0$. The last claim combines the relevant upper bound with $\tau_U \geq 1$. □

This formulation is intrinsic to the reduced capacity problem: it uses the two extreme coordinate classes of one zero-sum obstruction potential, rather than level sets from a three-color dual with redundant gauge freedom. Lemma 7.14 suggests a two-part proof: show that the two extreme sets of a support-orthogonal optimal potential are support-rigid, and then show that every such rigid extreme forces $\tau_U \geq 1$ for a positive circuit. By Lemma 7.15, the first part reduces further to proving that the support nullspace of a positive circuit is generated by support-twin differences.

The same argument gives a universal quantitative bound even when full recoloring fails. Define the *recoloring capacity* of a circuit to be the largest ρ for which the aggregate collection q can be decomposed into balanced colors with

$$\mu_{1,A} \geq \rho\lambda_{1,A}, \qquad \mu_{2,B} \geq \rho\lambda_{2,B}.$$

Corollary 7.18 (universal half-recoloring)**.** *Every positive circuit in a three-game star has recoloring capacity at least*

$$\rho_0 := \frac{q_A q_B}{\lambda_{1,A} q_B + \lambda_{2,B} q_A} \geq \frac{1}{2},$$

and the constant $1/2$ *is sharp. Moreover, on every balanced endpoint triple its Farkas margin* M_C *obeys the sharper bound*

$$M_C \geq (\rho_0 - 1)\big(\lambda_{1,A} d_A + \lambda_{2,B} d_B\big).$$

Proof. The aggregate q is balanced by Lemma 7.9. One valid coloring assigns all of q to game 1, and another assigns all of it to game 2. Put $a = q_A/\lambda_{1,A}$ and $b = q_B/\lambda_{2,B}$, and mix the first coloring with weight $b/(a+b)$. The retained fraction in each distinguished color is then

$$\frac{b}{a+b} a = \frac{a}{a+b} b = \frac{ab}{a+b} = \rho_0.$$

Since $q_A \geq \lambda_{1,A}$ and $q_B \geq \lambda_{2,B}$, we have $a, b \geq 1$ and hence $\rho_0 \geq 1/2$. Summing balancedness for these two colors yields

$$M_C \geq (\rho_0 - 1)\lambda_{1,A} d_A + (\rho_0 - 1)\lambda_{2,B} d_B,$$

which is the stated bound. For sharpness, use the Type-I orbit representative with $A = 7$, $B = 24$, and unit aggregate weights on $Q = \{5, 6, 7, 12, 19, 24, 25, 26\}$. Put

$$\pi_5 = \pi_7 = \pi_{25} = -\tfrac{1}{2}, quad \pi_6 = \pi_{26} = \tfrac{1}{2}$$

(all other $\pi_S = 0$), and

$$u_0 = u_1 = (\tfrac{1}{2}, -\tfrac{1}{2}, 0, 0, 0), \qquad u_2 = (1, 0, -\tfrac{1}{2}, \tfrac{1}{2}, -1).$$

These potentials have zero coordinate sum, and direct inspection gives

$$\pi_S + u_g(S) + \tfrac{1}{2}\mathbf{1}_{\{(g,S)=(1,A) \text{ or } (2,B)\}} \leq 0 \quad (g = 0, 1, 2, S \in Q),$$

while $\sum_{S \in Q} \pi_S = -1/2$. Multiplying by any recoloring weights and summing gives $\mu_{1,A} + \mu_{2,B} \leq 1$. Since both distinguished masses equal one, their common retained fraction is at most $1/2$. □

The two-level inequality and recoloring criterion in fact describe the full certificate class for a fixed circuit.

Proposition 7.19 (two-level certificate characterization)**.** *Fix the player set, two arms, and a both-sensitive positive circuit. Its Farkas right-hand side is nonnegative on every exact base-plus-two-bump triple if and only if its circuit inequality can be obtained by summing finitely many instances of Lemma 7.8, applied separately to the three games, together with nonnegative multiples of* $d_A \geq 0$ *and* $d_B \geq 0$*, after identifying their common base worths.*

Proof. The reverse implication follows by summing the valid inequalities. For the forward implication, add a common additive game to normalize the grand worth to one; this preserves exactness, the two bump sizes, and the circuit margin by stationarity. Introduce, for each game g and proper coalition U, a core allocation $x^{g,U}$ tight at U. These variables and the base worths form a finite linear system whose projection is exactly the set of exact endpoint triples. Maximize

the circuit-supported worth over this system. By hypothesis its optimum is at most the circuit efficiency total. Linear-programming duality supplies a dual optimum. Group its nonnegative core-inequality multipliers by the witness $x^{g,U}$. Cancellation of that witness's player coordinates against its efficiency and tightness equalities forces the group's coverage vector to be

$$\alpha_{g,U}\mathbf{1} + \delta_{g,U}\mathbf{1}_U.$$

Thus each group is an instance of Lemma 7.8; the only remaining bound multipliers are nonnegative multiples of the two bump inequalities. Their coefficient identity is the desired decomposition. □

An exact enumeration in Boolean linear rational arithmetic now replaces the earlier floating-point diagnostic. Here D has rank 12, so every circuit has at most 13 rows. Moreover, primitive circuit coefficients are minors of a $\{-1, 0, 1\}$ matrix; Hadamard's bound implies that, after normalizing their sum to one, every positive coefficient is at least $1/(13 \cdot 12^6) = 1/38{,}817{,}792$. Imposing this rational floor therefore loses no circuit. Exact single-game subcircuit cuts, arm-preserving orbit blocking, and terminal unsatisfiability at support 13 give the following finite result.

Proposition 7.20 (five-player two-arm theorem)**.** *Every three-game star* $\{b, b+d_Ae_A, b+d_Be_B\}$ *of exact games on at most five players is feasible.*

Proof. Corollary 7.5 and Proposition 7.29 leave, up to relabeling and interchange of the arms, only the disjoint five-player shapes

$$(|A|, |B|, |N \setminus (A \cup B)|) = (3, 2, 0) \quad \text{and} \quad (2, 2, 1).$$

For the first shape, exact enumeration through the rank bound yields 420 both-sensitive circuit orbits, or 4,914 labeled supports. Their support breakdown is

support size	10	11	12	13
orbits	2	24	83	311
labeled supports	18	270	954	3,672.

Of these, 411 orbits (4,812 supports) have an exact balanced recoloring with equality at the two distinguished rows. Eight more orbits (96 supports) have a recoloring with strict excess at one distinguished row, so Lemma 7.9 uses $d_A, d_B \geq 0$. The sole remaining orbit is the six-support Type-I orbit displayed above; the two singleton-tight exactness inequalities given there prove its circuit inequality for every exact endpoint triple.

For the second shape, terminal unsatisfiability occurs after 189 orbits (756 labeled supports): $9, 27, 153$ orbits of support $11, 12, 13$, respectively. Exact balanced recolorings neutralize 117 orbits (468 supports) with equality and the other 72 (288 supports) using bump nonnegativity. No exactness-specific exception occurs.

Every reported support has exact nullity one and a strictly positive rational kernel vector; every symmetry orbit and every neutralization identity is rechecked over $\mathbb{Q}$ by the standalone supplement. The determinant floor above and the terminal exact unsatisfiability queries establish that no positive circuit has been omitted. Hence every extreme Farkas ray using both changed-worth rows has nonnegative right-hand side. Proposition 7.6 and Lemma 7.7 then rule out infeasibility. □

The same certificate architecture persists in a nonexhaustive six-player frontier. Representative searches over all four disjoint shapes $(2,2,2)$, $(3,3,0)$, $(3,2,1)$, and $(4,2,0)$ expose 62 distinct exact circuit orbits, comprising 1,127 labeled supports. Their exact recoloring capacities range over $1/2, 2/3, 3/4, 1$. Every orbit with capacity below one nevertheless has endpoint optimum zero and a coefficientwise rational decomposition of the form in Proposition 7.19. This is discovery evidence, not an exhaustive $n = 6$ classification: its value is that no new certificate mechanism appears after the complete five-player theorem.

A second representative sweep covers all six disjoint arm shapes on seven players. It exposes 24 exact circuit orbits (1,365 labeled supports), of which 15 have recoloring capacity below one. Combining these with the five-player Type-I orbit and the 15 deficient six-player representatives gives 31 exact obstructions. In every case the reduced two-color capacity equals the original three-color capacity exactly. If z is the computed optimal potential in Proposition 7.13, then Corollary 7.11 applies with

$$U = P^+ = \operatorname{argmax}_i z_i \quad \text{or} \quad U = P^- = \operatorname{argmin}_i z_i.$$

Before quotienting the gauge, the maximum set succeeds in 29 cases and the minimum set in 29; the four failures stay exactly at ρ and are exactly the support-flexible coalitions identified by Lemma 7.14. After projecting each potential onto $K(q)^\perp$, all four disappear: both canonical extreme sets have capacity exactly one in all 31 cases, or 62/62 individual checks. Every maximum-set repair has $\delta < 0$, and every minimum-set repair has $\delta > 0$, matching the dual coordinate order. Thus the empirical existence problem has collapsed from arbitrary exactness decompositions to Conjecture 7.16, a canonical two-cut statement for a box-constrained degree zonotope. This remains discovery evidence, not a proof or an exhaustive seven-player classification. An independent exact rank audit supports the first proposed proof step: all 609 circuits in the exhaustive five-player census and all 108 representative or glued six- and seven-player circuits have twin-generated support nullspace. Fourteen have nonzero nullspace, always of dimension one; no non-twin direction occurs among the 717 circuits. The original census also suggested a sharp arm-ratio pattern. The scalar coloring $w = tq$ alone certifies capacity at least one for 97 circuits with neither arm exclusive: 73 lie exactly on its boundary and 24 satisfy it strictly. If $x = \lambda_{1,A}$, $y = \lambda_{2,B}$, $U = q_A - x$, and $V = q_B - y$, then the scalar condition is the product inequality $UV \geq xy$. All 97 additionally satisfy the stronger linear bounds $U \geq y$ and $V \geq x$.

Lemma 7.21 (One-row circuits glue local circuits)**.** *Suppose a support-minimal positive circuit contains exactly one monotonicity column on the A-edge and exactly one on the B-edge. In each of the three game blocks, the incident supported columns form a nullity-one local circuit.*

Proof. Write the two monotonicity columns as

$$m_A = (e_a, -e_a, 0), \qquad m_B = (e_b, 0, -e_b)$$

in the three game blocks, and let C_g be the matrix of supported core columns internal to block g. Each C_g has independent columns: otherwise a kernel vector supported only on those core columns would be a second global kernel direction. The positive global circuit restricts in the two leaf blocks to dependencies of $[C_1, -e_a]$ and $[C_2, -e_b]$. Adding one column to an independent set and obtaining a dependency makes each leaf nullity exactly one.

It remains to treat the root matrix $L_0 = [C_0, e_a, e_b]$. Every vector in $\ker L_0$ specifies two coefficients on e_a, e_b. The unique leaf relations can be scaled, with arbitrary real signs, to match those two coefficients, producing a unique global kernel vector. Conversely every global kernel

vector restricts to one in $\ker L_0$. Hence the global kernel is linearly isomorphic to $\ker L_0$. Its dimension is one by support minimality, so the root local support is also nullity one. □

Every circuit in the enlarged 717-record audit uses exactly one monotonicity row per arm, so Lemma 7.21 puts the entire observed frontier into this three-local-ray normal form.

Neither pattern is dimension-free. A targeted block-gluing construction produces the following exact six-player circuit outside the original representative sweep.

Proposition 7.22 (The scalar arm dichotomy fails)**.** *There is a support-minimal positive circuit with disjoint arms $A = \{1, 2, 3\}$ and $B = \{4, 5\}$ for which*

$$(x, y, U, V) = (1, 2, 1, 1).$$

Thus $U - y = -1$ and $UV - xy = -1$, so neither the linear cross-arm bounds nor the scalar product condition holds. Nevertheless its exact reduced two-color recoloring capacity is one.

Proof. Use one monotonicity column on each edge, for player 1 on the A-edge and player 4 on the B-edge. The core-row supports in games $0, 1, 2$ are, in bit-mask notation,

$$\{24, 3, 45\}, \qquad \{7, 58, 60\}, \qquad \{7, 24, 35, 37, 54\},$$

respectively. Give every row weight one except the game-2 row for coalition $24 = B$, which has weight two. In each game block the incident core and monotonicity columns form a nullity-one positive circuit, of support sizes $5, 4, 6$. Their shared monotonicity weights agree, so gluing the three local relations gives a 13-column global dependence. Direct rank calculation gives rank 12, hence the positive dependence is a circuit. Reading the four arm coefficients gives the displayed tuple.

The aggregate balanced collection has weights

$$q_3 = 1,\ q_7 = 2,\ q_{24} = 3,\ q_{35} = q_{37} = q_{45} = q_{54} = q_{58} = q_{60} = 1.$$

The subcollection

$$w_7 = w_{24} = w_{37} = w_{58} = 1$$

has degree two at every player, while $q - w$ is balanced of degree four. It retains one copy of A and two copies of B, so it repairs both full bumps at capacity one. The companion exact LP dual gives the matching upper bound. All ranks, stationarity equations, and primal–dual identities are checked by the supplement verifier. □

The original 31 deficient circuits all have an exclusive arm, and their 62 canonical extreme witnesses attain the matching ceiling one. The local-gluing normal form makes it possible to search beyond that frontier systematically. It also falsifies the stronger direct-recoloring target.

Proposition 7.23 (Nonexclusive balanced recoloring can fail)**.** *There is a support-minimal positive circuit with $A = \{1, 2, 3\}$ and $B = \{4, 5\}$, neither arm exclusive, whose ordinary recoloring capacity is $3/4$. For an optimal support-orthogonal potential, however, its maximum and minimum coordinate sets have fixed-witness capacities $5/4$ and $3/2$, respectively.*

Proof. Use the same two monotonicity columns as in Proposition 7.22, with weights two on the A-edge and one on the B-edge. The three core supports and weights are

$$\begin{array}{c|l} 0 & (3,1),(5,1),(24,1),(41,1) \\ 1 & (7,2),(26,2),(28,2),(32,4) \\ 2 & (7,1),(24,2),(35,1),(37,1),(54,1). \end{array}$$

Each block is a local circuit and the shared edge weights match. The global support has 15 columns and rank 14, so it is a positive circuit. Its arm masses are

$$(x, y, U, V) = (2, 2, 1, 1).$$

Exact primal–dual solution of the reduced box system gives $\rho = 3/4$. One dual optimum has

$$z = \frac{1}{4}(1, -1, -1, -1, 2, 0),$$

whose maximum and minimum sets are $\{5\}$ and $\{2, 3, 4\}$, with masks 16 and 14. Imposing those sets in the fixed-witness system gives exact primal–dual values $5/4$ and $3/2$. The machine-readable support, subcollections, potentials, and bound identities are supplied in the supplement. □

The expanded curated audit now contains 717 circuits: 52 have ordinary capacity below one, of which 21 are nonexclusive glued circuits. The factorized search exactly solves 5,035 selected capacity problems, and both canonical extreme witnesses repair all 21 deficient cases. Their ordinary capacities are $3/4$, $5/6$, $7/8$, or $11/12$, while their fixed-witness capacities range from $11/10$ to $3/2$. Thus exactness-specific two-level repair is genuinely indispensable, and Conjecture 7.16 survives a substantially more adversarial frontier than balanced recoloring alone.

A further exact audit fixes the same singleton witness $\{5\}$ (mask 16) across all 59,304 product-forbidden gluings in this factorized local pattern, rather than only the 21 deficient circuits in the selected ordinary-capacity screen. Every fixed-witness capacity is strictly above one, and the sharp bound is $9/8$, attained by 45 circuits. The spectrum has 21 values; the returned exact optima use 19 projective dual rays and 18 potential weak-order types, with the solver selecting two rays on the minimum frontier. An independent exact enumeration of the complete dual hyperplane arrangement gives 87 feasible projective rays. Four are sharp, in three symmetry orbits, and all $87 \cdot 59{,}304 = 5{,}159{,}448$ ray–circuit evaluations satisfy the $9/8$ bound. Arrangement minimization reproduces the complete LP capacity histogram exactly. The local enumeration uses exact rank modulo 61—valid because Hadamard's bound puts every local minor below 56 in absolute value—followed by signed-integer-cofactor kernels. Thus the $9/8$ statement is exhaustive for this specified gluing pattern. The exhaustive table also factors locally. For each arrangement ray, the cross-multiplied margin is a root score plus the two matched root weights times independently minimized leaf rates. Checking the 626 first leaves, three second leaves, and 56 roots for all 87 rays requires 59,595 exact local comparisons; no local margin is negative, and the only four zero rows are the globally sharp rays. Because the local and arrangement coalition universes coincide, this proves the same $9/8$ bound for all $56 \cdot 626 \cdot 3 = 105{,}168$ raw gluing triples in the pattern, not only the product-forbidden subfamily. Positive-part subadditivity compresses the evaluation stage further: the first leaf has a closed degree-only bound, and the roots have only 12 relevant mass types. These bounds prove 86 of the 87 rays; the remaining ray needs exact incidence checks at only three roots, with margins $10, 2, 2$. Thus the final local

evaluation uses $87 \cdot 12 + 3 = 1{,}047$ comparisons. It remains a finite-pattern theorem, not a proof of Conjecture 7.16 outside that pattern.

A natural attempt to remove the finite first-leaf enumeration is false. The bound $q_A \le |A| m_A$, where m_A is the matched singleton coefficient, holds for all 2,315 local circuits through six players but fails under the standard new-player construction for minimal balanced collections (Laplace Mermoud et al. 2025). A complete implementation starts from all 1,292 five-player minimal balanced collections and reproduces the published total of 200,214 six-player collections. Extending those sources yields 1,320 seven-player leaves with a three-player arm ratio five. One representative is

$$(\{1,2\},\{1,3\},\{2,3,4\},\{1,4,5\},\{1,4,6\},\{7\},\{5,6,7\}),$$
$$6\lambda = (2,2,4,1,1,1,5).$$

Its incidence columns have rank seven, so it is minimal, and $q_{\{5,6,7\}}/m_{\{7\}} = 5$. Gluing this leaf to lifted copies of all 56 roots and all three second leaves gives 168 exact nullity-one circuits; 21 have fixed-$\{3\}$ capacity below 9/8, with exact minimum 13/12. The 1,320 leaves form four distinguished-arm symmetry orbits. Auditing one canonical representative of each checks 672 global circuits: every orbit breaks 9/8, but none breaks one. Hence the fixed six-player 9/8 theorem has no dimension-free extension. The unit repair threshold relevant to Conjecture 7.16 survives this stronger frontier.

The survival has a structural explanation that removes the first-leaf enumeration entirely. Write q^0, q^1, q^2 for the root and two leaf core collections after matching coefficients $m_A, m_B > 0$, and let a, b be the selected players. Thus

$$\deg(q^0) = \alpha_0 \mathbf{1} + m_A e_a + m_B e_b, \quad \deg(q^1) = \alpha_1 \mathbf{1} - m_A e_a, \quad \deg(q^2) = \alpha_2 \mathbf{1} - m_B e_b.$$

Proposition 7.24 (Unit root absorption)**.** *Fix a proper witness U. If there is a subcollection $z \le q^0 + q^2$ with*

$$\deg(z) = \beta \mathbf{1} + m_A e_a + \delta \mathbf{1}_U, \qquad z_B \le q^0_B,$$

then the fixed-U capacity of the glued circuit is at least one, for every compatible first leaf q^1 and without any bound on q^1_A/m_A.

Proof. Put $w = q^1 + z$. Then $0 \le w \le q^0 + q^1 + q^2$ and $\deg(w) = (\alpha_1 + \beta)\mathbf{1} + \delta \mathbf{1}_U$. It retains the full first requirement because $w_A \ge q^1_A$, while

$$(q - w)_B = q^0_B + q^2_B - z_B \ge q^2_B,$$

which retains the full second requirement. □

For the lifted pattern above, all $56 \cdot 3 = 168$ root–second-leaf pairs admit such a subcollection in exact rational arithmetic. For a dual ray (r_A, r_B, z) from Proposition 7.13, let

$$R = r_B q^0_B + \sum_T (q^0_T + q^2_T)[z(T) + r_A \mathbf{1}_{\{T=A\}} - r_B \mathbf{1}_{\{T=B\}}]_+ - m_A z_a.$$

An independent dual audit gives the stronger reserve

$$R(q^0, q^2) \ge \frac{1}{3}(m_A r_A + m_B r_B)$$

for every feasible fixed-witness dual ray. Since the three second leaves have $q_B^2/m_B = 2$, this implies the first-leaf-uniform estimate

$$\tau_U \geq 1 + \frac{1}{3\max\{q_A^1/m_A, 2\}}.$$

In particular the ratio-five leaves satisfy $\tau_U \geq 16/15$. This is weaker than their exact minimum $13/12$, but it proves capacity one for arbitrary compatible first leaves, not only the four enumerated orbits. The remaining finite input has moved entirely to the root–second-leaf pair. As an exact stress test, all four Peleg constructions from the 56 regularized roots produce 604 distinct seven-player roots. All $604 \cdot 3 = 1{,}812$ pairings with the second leaves admit absorbers, and the normalized dual reserve still has minimum $1/3$ in every Peleg case. This points to Peleg-closure of root absorption as the next structural lemma. The first Peleg case can already be closed. If I is its marked subcollection, the interpolation equations $v(T) = \mathbf{1}_{\{T \in I\}}$ on the regularized root imply $v(N) = 1$ and $v_a = v_b = v_u = 0$: use $\lambda(I) = 1$, the two marked cosingletons, and the unmarked coalition $B = \{b, u\}$. A root-only absorber therefore lifts directly, because its marked mass is exactly its baseline degree. This covers 93 markings. For the remaining 66, arising from the 23 non-root-only parents, the interpolation potentials have only five types and are nonnegative on the complete second-leaf support. Hence $[x + sv(T)]_+ \leq [x]_+ + v(T)[s]_+$ transports the parent dual remainder to the child; all 159 admissible case-1 markings preserve unit absorption, and the adversarial 66 preserve the $1/3$ reserve. Case 2 also closes. If κ is the new-singleton weight, the corrected potential $\widetilde{v} = v + \kappa(e_a + e_b - e_u)$ satisfies $\widetilde{v}(N) = 1$ and $\widetilde{v}_a = \widetilde{v}_b = \widetilde{v}_u = 0$. Its correction has total root mass one and total second-leaf mass α_2, so the two new singleton hinges pay the positive-part errors exactly. The required support signs hold in all 113 admissible emissions. Case 3 also closes, but requires a role-specific split argument. For split coalition D, weight λ_D, and $\theta = \kappa/\lambda_D$, assign interpolation value θ to D, one to marked columns, and zero otherwise. Convexity then transports the split column exactly. The 104 emissions divide into 21 $\bar{a}$-splits, 62 $\bar{b}$-splits, and 21 B-splits. The $\bar{a}$ role has a nonnegative second-support potential and its changed coefficient leaves the hinge $(m_A - \kappa)([z(\bar{a})]_+ - z(\bar{a})) \geq 0$. For the B role, the witness correction $\widetilde{w} = w + \theta(e_b - e_u)$ is nonnegative on both relevant supports; the sole split-hinge loss is at most κr_B, and exact primal–dual certificates prove the sharp parent inequality $R \geq \kappa r_B$ in all 21 parent–second pairings. Finally, 50 of the 62 $\bar{b}$-split children absorb from the root alone. For the remaining 12, a split-aware absorber keeps root and second allocations separate and uses the private new singleton to make the new-player degree equal the old baseline; all $12 \cdot 3 = 36$ exact systems are feasible. Hence cases 1–3 preserve unit root absorption on the certified family. Case 4 now closes by the same split-aware primal principle. If $\omega = (1-t)\mu + t\nu$ is its mixed parent collection and $\omega(I) = 1$, keep source-root and second-leaf allocations z^0, z^2 separate and choose private-singleton mass s with

$$\deg(z^0 + z^2) = \beta\mathbf{1} + m_A e_a + \delta e_u, \qquad z^0(I) + s = \beta, \qquad (z^0 + z^2)_B \leq q_B^{\omega}.$$

Marking z^0, retaining z^2, and adding s of the new singleton then gives the child absorber: the new-player degree is exactly β. Among 324 rank-one parent unions, 201 pairs yield 228 distinct decoded emissions. All $228 \cdot 3 = 684$ source systems are feasible in exact rational arithmetic, and all mapped child certificates verify coefficientwise. Thus all four Peleg constructions preserve unit root absorption on the certified family.

Proposition 7.25 (Disjoint arms: exactness chaining)**.** *Let b be exact and $t_A = b + d_A e_A$, $t_B = b + d_B e_B$ exact, where A, B are nonempty proper coalitions with $A \cap B = \emptyset$ and $R :=$*

$N \setminus (A \cup B) \neq \emptyset$. *Then*

$$b(N) \;\geq\; b(A) + d_A + b(B) + d_B + b(R),$$

i.e. the partition Farkas obstruction $\{A, B, R\}$ *to the balancedness of* $M = b + d_A e_A + d_B e_B$ *vanishes. (This is one obstruction among many; the proposition does not by itself establish balancedness of* M *or feasibility of the family.)*

Proof. Step 1. Every $x \in \mathrm{Core}(t_B)$ has $x(N \setminus A) = x(B) + x(R) \geq b(B) + d_B + b(R)$; since t_B is exact at $N \setminus A$ and $t_B(N \setminus A) = b(N \setminus A)$ (here $B \subsetneq N \setminus A$ because $R \neq \emptyset$, so the bump at B does not change the worth of $N \setminus A$), taking the minimum gives $b(N \setminus A) \geq b(B) + d_B + b(R)$. *Step 2.* By exactness of t_A at $N \setminus A$ there is $x^* \in \mathrm{Core}(t_A)$ with $x^*(N \setminus A) = b(N \setminus A)$, hence $x^*(A) = b(N) - b(N \setminus A)$; but $x^*(A) \geq b(A) + d_A$, so $b(N) \geq b(A) + d_A + b(N \setminus A)$. Chaining the two steps gives the claim. □

These techniques have a genuine limit, which the following example (due to a reviewer of an earlier draft) makes precise: common-core intersection of bumped games—the conclusion of Lemma 7.3—can fail when $m = n = 4$ arms are bumped, even though the associated monotonicity family remains feasible.

Proposition 7.26 (Core intersection can fail at $m = n = 4$; feasibility survives)**.** *Let* $n = 4$, $b(N) = 5$ *and* $b(S) = 0$ *for all proper* S. *For* $k \in N$ *let* $t_k = b + 4\, e_{N\setminus\{k\}}$. *Then* b *and each* t_k *are exact, and* $\mathrm{Core}(t_k) = \{x \geq 0 :\ x(N) = 5,\ x_k \leq 1\}$, *so* $\bigcap_k \mathrm{Core}(t_k) = \emptyset$ *(four inequalities* $x_k \leq 1$ *would force* $x(N) \leq 4$*); equivalently, the all-arm game* $b + \sum_k 4\, e_{N\setminus\{k\}}$ *is unbalanced. Nevertheless the star family* $\{b, t_1, \ldots, t_4\}$ *is feasible: take* $\theta(b) = (2, 1, 1, 1)$, $\theta(t_1) = (1, \frac{4}{3}, \frac{4}{3}, \frac{4}{3})$, *and for* $k \neq 1$ *the point of* $\mathrm{Core}(t_k)$ *obtained from* $(2, 1, 1, 1)$ *by moving one unit from player* k *to player* 1 *(e.g.* $\theta(t_2) = (3, 0, 1, 1)$*), which dominates* $\theta(b)$ *on* $N \setminus \{k\}$.

Proof. Exactness of b is clear (all mass can be placed on any single player). For t_k: the core is as displayed. For a proper $S \ni k$, the point with $x_k = 0$ and mass 5 on a player outside S attains $x(S) = 0 = t_k(S)$. For a proper $S \not\ni k$ with $S \neq N \setminus \{k\}$, there is a player $j \notin S \cup \{k\}$; the point with $x_k = 1$, $x_j = 4$ attains $x(S) = 0 = t_k(S)$. Finally the minimum of $x(N \setminus \{k\}) = 5 - x_k$ over the core is $4 = t_k(N \setminus \{k\})$. So t_k is exact. Emptiness of the intersection and unbalancedness (the collection $\{N \setminus \{k\}\}_k$ with weights $1/3$ gives $\frac{16}{3} > 5$) are immediate. Feasibility: each displayed $\theta(t_k)$ lies in $\mathrm{Core}(t_k)$ and dominates $\theta(b) = (2, 1, 1, 1)$ on every coordinate of $N \setminus \{k\}$. □

Proposition 7.26 corrects a conjecture we advertised in an earlier draft (that the all-arm game of exact single bumps is always balanced): the common-point method cannot decide the exact-games question, because family feasibility is strictly weaker than core intersection. Computationally, across more than 10^6 family-feasibility linear programs built from over 7,500 exact four-player bases (two- and three-arm stars and depth-two lattices), floating-point screening flagged *no infeasible family of exact games*. The pipeline certifies any flagged infeasibility by an exact Farkas certificate (validated on the Housman–Clark family and on Theorem 3.2); the negative outcome itself is float-screen-level, and float screening can in principle produce false negatives near degeneracy. In light of Proposition 7.29 below, the four-player census outcome is in fact implied by the existence of a monotonic selection on $\mathcal{E}_4$, so we present it as validation of the feasibility pipeline rather than as evidence at the open frontier, which begins at $n = 5$.

7.3 Equivalence, and the state of the question

Proposition 7.27. *Fix n and let $\mathcal{E}_n$ be the class of exact n-player games. A coalitionally monotonic core selection exists on $\mathcal{E}_n$ if and only if every finite bump family within $\mathcal{E}_n$ is feasible.*

Proof. Necessity is immediate. For sufficiency, consider the product $P = \prod_{v \in \mathcal{E}_n} \mathrm{Core}(v)$, compact in the product topology by Tychonoff's theorem, since each core is a nonempty compact subset of $\mathbb{R}^n$. For each bump pair (u, v) in $\mathcal{E}_n$ let $C_{u,v} \subseteq P$ be the set of selections satisfying the (finitely many, closed) monotonicity inequalities of Definition 2.1 for that pair; $C_{u,v}$ is closed and depends on two coordinates only. A finite intersection of such sets involves finitely many games; feasibility of the corresponding finite family provides a point (extending arbitrarily elsewhere). By the finite intersection property the full intersection is nonempty, and any of its points is a coalitionally monotonic core selection. □

Proposition 7.28 (structure of a minimal finite obstruction)**.** *Suppose a finite bump family on q games is infeasible and is inclusion-minimal with respect to its games and bump relations. Label each directed bump edge by its bumped coalition. Then:*

(i) the underlying undirected graph is connected and the directed graph is acyclic;

(ii) for every edge $u \to w$ labeled S, some $i \in S$ has no alternative directed path from u to w all of whose edge labels contain i;

(iii) infeasibility is witnessed by at most $q(n-1)+1$ core or monotonicity halfspaces, after the q efficiency equalities are imposed. For rational games, the corresponding sparse Farkas certificate may be chosen rational.

Proof. If the undirected graph were disconnected, feasible selections for its components could be combined, contrary to minimal infeasibility. Along every directed edge exactly one coalition worth strictly increases, so the sum of all coalition worths is a strictly increasing potential; directed cycles are impossible.

For (ii), suppose instead that for every $i \in S$ there were such an alternative path. Transitivity of the monotonicity inequalities along that path would imply $\theta_i(u) \le \theta_i(w)$. Thus every inequality attached to the edge $u \to w$ would follow from the remaining edges, contradicting minimality.

Finally restrict the qn allocation variables to the product of the q efficiency hyperplanes, an affine space of dimension $q(n-1)$. Family feasibility is the intersection there of finitely many core and monotonicity halfspaces. Helly's theorem gives an empty subintersection of at most $q(n-1)+1$ halfspaces. With rational coefficients, the usual rational form of Farkas' lemma supplies rational multipliers on such a subfamily. □

Proposition 7.28 turns the search for a refuting exact family into a sparse labeled-DAG problem. In particular, disconnected stars, directed cycles, transitively redundant bump labels, and certificates whose support exceeds the Helly bound need not be searched when mining minimal obstructions.

For $n \le 4$ the existence question is in fact already settled in the affirmative by combining two known results:

Recall that the core of v is *large* (Sharkey 1982) if for every $y \in \mathbb{R}^N$ with $y(S) \ge v(S)$ for all S there is $x \in \mathrm{Core}(v)$ with $x \le y$, and call a game *PES-stable* if the procedural egalitarian solution of Dietzenbacher (2021) selects a core allocation of it.

Proposition 7.29 (Dietzenbacher 2021, with Biswas et al. 1999)**.** *A coalitionally monotonic core selection exists on the class $\mathcal{E}_4$ of exact four-player games (hence also on $\mathcal{E}_n$, $n \leq 4$).*

Proof. Every exact game with at most four players has a large core (Biswas et al. 1999). Large-core games are PES-stable, and on PES-stable games the procedural egalitarian solution is a single-valued core selection satisfying coalitional monotonicity (Dietzenbacher 2021); the combination is observed in Dietzenbacher (2021). (As a consistency check, we verified by exact vertex enumeration that all eight exact four-player games directly certified in this paper have large cores.) □

Proposition 7.27 frames both sides of the general ledger: the impossibility on balanced games *is* one infeasible finite family (Theorem 3.2); the results above are partial progress toward feasibility of all finite exact families for general n. The solutions studied in this paper fail coalitional monotonicity on exact games individually—the nucleolus (already on convex games (Hokari 2000)), the per-capita nucleolus (Arin 2013), the core-center (Mirás Calvo et al. 2021) (and, in our tests, a coordinate-extreme variant, in 136 of 1,059 exact bumps), and the SD-prenucleolus (Theorem 5.1)—while Proposition 7.29 shows that for $n \leq 4$ a different, egalitarian-type formula succeeds. For $n \geq 5$ no selection is known to work. Dietzenbacher's particular procedural egalitarian solution cannot supply one on all exact games: Mauri Masdeu (2017, Example 5) applies the same maximum-average procedure to an exact five-player game, obtains the zero vector, and observes that it lies outside the core. The recent follow-up of Dietzenbacher and Doğan (2025) develops egalitarianism under monotonicity constraints further but does not settle the existence of a different selection on exact games.

Conjecture 7.30. *For every fixed $n \geq 5$, a coalitionally monotonic core selection exists on the class $\mathcal{E}_n$ of exact n-player games.*

By Proposition 7.27, Conjecture 7.30 is equivalent to feasibility of all finite exact families for the given n. The pinning-based and partition-aggregated obstruction mechanisms are unavailable at any n (Lemma 7.1, Theorem 7.2); what is missing is a feasibility proof for arbitrary finite families—or a refuting family. Proposition 7.26 shows that one natural route to refutation, empty common intersection of bumped cores, is by itself insufficient. An argument for arbitrary finite families—or a new finite obstruction—is therefore still required.

8 Verification methodology

All results were produced by a search-then-certify discipline. Screening uses floating-point successive linear programs; every reported object is then reconstructed in exact rational arithmetic and certified independently of the search path: nucleolus and SD points by Kohlberg-type balanced-collection criteria evaluated by an exact rational simplex method (written for this project after we found that a current computer-algebra system's exact linear-programming routine returns infeasible points as optima on three-variable instances; our solver is fuzz-tested against a standard floating-point solver with exact feasibility assertions), class memberships (balancedness, total balancedness, exactness, convexity) by exact linear programs, family infeasibility by exact Farkas certificates re-verified under constraint reordering, and headline witnesses additionally re-derived by an algorithmically independent exact successive-LP solver. Negative search outcomes (no violation found, no infeasible family found) are reported at the float-screen level: exact certification in those campaigns is triggered by candidate positives, and none arose;

the sole fully exact negative sweep is the exhaustive 2×2 assignment grid. Known values from the literature (Talmud division, serial rule, minimal-overlap rule, a published five-player nucleolus, Hokari-type failures at their known minimal sizes, and the Housman–Clark family itself) are reproduced as anchors. The discipline caught nine false or overstated candidate results before any claim was circulated, including two floating-point artifacts near the exactness boundary and the overreaching core-intersection conjecture corrected in Proposition 7.26; we accordingly treat direct exact LP oracles as authoritative over derived closed forms. The computational search, the verification tooling, and the drafting of this paper made substantial use of large-language-model assistance; every mathematical claim was verified by the certification pipeline described here, and all text was reviewed by the author. Computations used Python 3.10 with exact arithmetic via the standard `fractions` module and floating-point screening via NumPy 2.2/SciPy 1.15 (HiGHS). A machine-readable supplement (`certificates.json`) contains, for every computational theorem, the full game tables, per-coalition exactness witnesses, complete satisfaction/excess tables, and strictly positive balancing weights for every cumulative level set, together with a one-command verifier (`verify_paper.py`) that re-checks all of it in exact arithmetic using only the Python standard library (it recomputes every satisfaction and excess table from scratch, checks that every stored level set equals the recomputed one, and validates the supplied balancing weights by pure rational arithmetic), and an environment manifest; together these form Online Resource 1.

9 Open problems

1. Prove (or refute) feasibility of every finite bump family of exact n-player games for $n \geq 5$; by Proposition 7.27 this settles Conjecture 7.30. Proposition 7.26 shows the common-core-point method cannot decide it, and the procedural egalitarian route is closed by the exact counterexample of Mauri Masdeu (2017, Example 5). For the disjoint two-bump frontier, the sharp intermediate target is to prove Conjecture 7.16: both extreme coordinate classes of a support-orthogonal optimal potential must make the box system in Corollary 7.11 feasible.

2. Recover the unpublished Solymosi–Brugueras–Raghavan proof (see Section 6) that the nucleolus is pairwise monotonic on assignment games, and prove Conjecture 6.2 for the SD-prenucleolus. Theorem 6.1 shows two-sidedness must be used.

3. Prove Conjecture 6.3 (Dutta–Kar cost monotonicity of the nucleolus on MCST games), known on the fixed-tree subclass (Maschler et al. 2010).

4. Characterize SD-relevance (Arin and Katsev 2018) within the exact games; Theorem 5.1 shows exactness is insufficient.

5. Determine whether the witness of Theorem 6.1 is pre-empted by an example in the introduction of Miquel (2009).

Statements and Declarations

Funding: none. **Data and code availability:** a machine-readable certificate supplement accompanies this paper as Online Resource 1 (`certificates.json`; `verify_paper.py`; `environment.txt`); the verifier is self-contained and uses only the Python standard library. **Use of AI assistance:** see Section 8. **Competing interests:** none.

A Certificates

All certificates below are exact rational data; balancing weights are strictly positive multipliers λ with $\sum_S \lambda_S \mathbf{1}_S = \mathbf{1}_N$. This appendix displays the key tables; the *complete* certificate package—per-coalition exactness witnesses for the eight games claimed exact by direct certification (the convex game of Theorem 5.4 is exact by convexity), full satisfaction and excess tables, and balancing weights for *every* cumulative level set of every solution point claimed (including the SD points of the permutation games and the two-player subgame of Theorem 5.4)—is in the machine-readable supplement (`certificates.json`), re-verifiable in exact arithmetic, using only the Python standard library, with the single command `python3 verify_paper.py`.

Theorem 5.1

Satisfaction levels of SD(v) (ascending; coalitions at each level):

$$\begin{aligned}
\alpha = 15/4: &\quad \{3\},\ \{2,3\},\ \{1,2,4\},\ \{1,3,4\}\\
\alpha = 53/12: &\quad \{1\},\ \{1,3\},\ \{1,2,3\},\ \{2,4\}\\
\alpha = 19/4: &\quad \{2\},\ \{1,2\}\\
\alpha = 61/12: &\quad \{2,3,4\}\\
\alpha = 205/12: &\quad \{4\},\ \{1,4\},\ \{3,4\}
\end{aligned}$$

The minimal level set is balanced with weights $\lambda_{\{3\}} = \frac{1}{3}$, $\lambda_{\{2,3\}} = \frac{1}{3}$, $\lambda_{\{1,2,4\}} = \frac{2}{3}$, $\lambda_{\{1,3,4\}} = \frac{1}{3}$. For SD($v'$) the levels are

$$\begin{aligned}
\alpha = 29/8: &\quad \{2,3\},\ \{1,2,4\},\ \{1,3,4\}\\
\alpha = 33/8: &\quad \{2\},\ \{3\}\\
\alpha = 35/8: &\quad \{1\},\ \{1,2\},\ \{1,3\},\ \{1,2,3\},\ \{2,4\}\\
\alpha = 43/8: &\quad \{2,3,4\}\\
\alpha = 139/8: &\quad \{4\},\ \{1,4\},\ \{3,4\}
\end{aligned}$$

with minimal level set balanced by $\lambda_{\{2,3\}} = \lambda_{\{1,2,4\}} = \lambda_{\{1,3,4\}} = \frac{1}{2}$. Cumulative lower level sets are balanced at every threshold (weights in the supplement), certifying both points by the criterion of Section 4.

Theorem 5.3

At every sampled point of the segment ($t = -\frac{1}{2}, -\frac{1}{3}, -\frac{1}{4}, 0, \frac{1}{4}, \frac{1}{3}, \frac{1}{2}$, all in the supplement) the bilateral complaints are

$$s_{12} = s_{21} = 2, \qquad s_{34} = s_{43} = 2, \qquad s_{ij} = 0 \text{ for all other ordered pairs},$$

and the two half-interval stage orders displayed in the proof of Theorem 5.3 extend these equalities to the whole segment.

Theorem 6.1

For the base game K: the nucleolus $(\frac{1}{2}, \frac{33}{4}, \frac{1}{2}, \frac{7}{4})$ has maximal excess 0, attained exactly by the collection $\{\{1,3\},\{2,4\}\}$, balanced with weights $\lambda = 1, 1$ (a partition). For the bumped game K': the nucleolus $(\frac{2}{3}, \frac{23}{3}, \frac{1}{3}, \frac{7}{3})$ likewise has maximal excess 0 attained by the partition $\{\{1,3\},\{2,4\}\}$. Kohlberg's criterion requires every cumulative upper excess set to be balanced; the supplement lists all eight (resp. seven) thresholds with strictly positive weights for both games, together with the complete SD certificates for the SD subclaim.

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